[DRAFT] Provision of Airspace Conditions in Geozone format as part of Skyguide ANSP CIS provisionSWIM Service Description Document

General Service Description	
Service description identification **SWIM-SERV-030**	**Title:** Provision of Airspace Conditions in Geozone format as part of Skyguide ANSP CIS provision SWIM Service Description Document **Edition:** V0.1 **Publication Date:** TBD
Service identification **SWIM-SERV-040**	**Name:** Provision of Airspace Conditions in Geozone format as part of Skyguide ANSP CIS provision SWIM Service **Version:** V1.0
Service abstract **SWIM-SERV-050**	Skyguide's obligations in the provision of airspace conditions as part of CIS are defined in Skyguide's document "Concept of Operation / Airspace Conditions to CIS". As defined in Skyguide's Concept of Operation, all airspace conditions relevant to be provided through CIS will be provided in the format of a geozone. To comply with the requirements defined in Skyguide's Concept of Operation, the service implementation is based on the ASTM F3548-21 standard, already identified as an applicable standard in U-space accepted means of compliance and guidance material in the frame of the UAS Flight Authorisation.
Service Provider and Contact Information	
Service Provider **SWIM-SERV-060**	**Name:** Skyguide **Description of organization:** Swiss ANSP
Service Characteristics	
Geographical extent of	Geographical coverage of exchanged information service payload is available through the DSS under ASTM Section A2.

information **SWIM-SERV-090**	Unless specified differently in the geozone itself, Skyguide provides geozones valid within the Swiss sovereign airspace.
Service categories **SWIM-SERV-100**	**Service category:** Airspace Conditions
Service lifecycle information **SWIM-SERV-110**	**Lifecycle stage and if applicable qualifying dates:** Until further notice
Service standard reference **SWIM-SERV-120**	**Service standard reference:** **1.** Technical Specifications for Geo Zones and U-Space Data provision and exchange (ED-318) **Implemented options of the service standard:** ● Information Retrieval and Update Service is implemented, using ASTM F3548 constraint management ○ the DSS subscription mechanism ○ The Retrieval interface ○ The Publication interface **2.** Standard Specification for UAS Traffic Management (UTM) UAS Service Supplier (USS) Interoperability (ASTM F3548-21) **Implemented options of the service standard:** ● Skyguide act as authorised constraint provider role ● USSPs are expected to implement the constraint processor role ● Requirements in Section A2.4 regarding down USS not used ● Certain parameters in section A1 are extended. ● USS-USS interface amended to add ED318 as a parameter (https://github.com/skyguide-ansp/cis-interfaces) Discovery and synchronization service configuration process - as defined by InterUSS DSS Pooling instructions on joining an existing pool with new instance.
High-level Description of Service Offer	
Operational environment **SWIM-SERV-130**	Refer to the following documents for operational needs addressed by the service and the capabilities offered by the service. ● Skyguide's Concept of Operation ● Commission Implementing Regulation (EU) 2021/664 - Preamble ● ASTM F3548-21 Section A2 capabilities include: - USS-DSS Interfaces - Other DSS Interfaces - USS-USS Interfaces (proposed update with ED318)

	- DSS-DSS Synchronization
Service functions **SWIM-SERV-140**	Refer to the following documents for functions offered by the service and their real world effects. • Skyguide's Concept of Operation • Commission Implementing Regulation (EU) 2021/664 Article 5 • Article 5 Accepted Means of Compliance and Guidance Material to Regulation (EU) 2021/664 — Issue 1 • ED318 Chapter 2, Data Scope • ASTM F3548-21 Section A2 functions are provided for the following capabilities: - USS-DSS Interfaces - Other DSS Interfaces - USS-USS Interfaces (proposed update with ED318) - DSS-DSS Synchronization
Limitations and Constraints on Using the Service	
Service access and use conditions **SWIM-SERV-150**	Service access and use conditions are prescribed by FOCA, as required by Regulation (EU) 2021/664, Art. 3 (4), as part of the U-space services performance requirements.
Security constraints **SWIM-SERV-160**	ASTM F3548-21 Sections 5.2 and A2 (relevant requirements are listed below): * GEN0005 * DSS0200 * DSS0205 *Per ASTM F3548-21, Appendix X1 - Reference architecture for interoperability security controls, The X1.1 Base Deployment: Access Tokens with Audience Claims is implemented. Authentication and Authorization for communication between USS-USS and USS-DSS shall be facilitated by a common interoperability token exchanger service supporting OAuth 2.0.*
Quality Aspects	
Quality of service **SWIM-SERV-180**	Skyguide's Concept of Operation defines various data and service quality requirements. In regards to service availability, for the U-space V1, the service is provided with a 99% availability.
Service validation information	Service will be verified using the InterUSS Automated Test Baseline prior to software deployment using test baseline identifier XXXX.

<table>
<tr><td>SWIM-SERV-200</td><td></td></tr>
<tr><td colspan="2">Behavior of the Service</td></tr>
<tr><td>Application message exchange pattern

SWIM-SERV-210</td><td>Refer to the following sections in ASTM F3548-21 for information about the application message exchange pattern(s) used by the service.
* A2 - Interoperability requirements and DSS testing
* A3 - USS-DSS and USS-USS OpenAPI YAML Description proposed with ED318</td></tr>
<tr><td>Service behavior

SWIM-SERV-220</td><td>Service behaves as constraint manager with the provider role in ASTM F3548-21.
However certain limitations have been amended from section A1, namely :
- CstrMaxDuration extended from 24h to 1year due to the publication of static information, republished at every AIRAC cycle .

The publication of permanent geozones is renewed at every AIRAC cycle with a validity truncated at CstrMaxDuration.

For Uspace V1, the following type of aeronautical data will be included :
<table>
<tr><th>Airspace Type</th><th>Description</th><th>Remark</th></tr>
<tr><td>ATS Airspace</td><td>Control Zone (CTR)</td><td>/</td></tr>
<tr><td rowspan="3">Special Activity Airspace</td><td>Prohibited Area</td><td>Only when the lower limit is below 3 000 ft GND</td></tr>
<tr><td>Restricted Area</td><td>Only when the lower limit is below 3 000 ft GND</td></tr>
<tr><td>Danger Area</td><td>Only when the lower limit is below 3 000 ft GND</td></tr>
<tr><td rowspan="4">Other activities of a dangerous nature and other potential hazards</td><td>Artillery firing areas</td><td>Only relevant for ground risk.</td></tr>
<tr><td>Restricted Areas for Gliders</td><td>Only when the lower limit is below 3 000 ft GND</td></tr>
<tr><td>Glider Sectors</td><td>Normally, with a CTR</td></tr>
<tr><td>Anti-hail Rocket Firing Area</td><td>Only relevant for ground risk.</td></tr>
</table>

For controlled aerodromes without specified operational hours (Hx), the CTR exists only during the ATC operational hours. Geozone related to CTR is, therefore, unpublished outside ATC operational hours.
CTR will be re-published with a provisional start-time corresponding to predefined aerodrome opening hours (not to be mistaken for ATC operational hours) published in the AIP Switzerland. If ATC</td></tr>
</table>

	operational hours start before the provisional start time originally published, the start time will be updated. For other airspace restrictions, the following cases are considered: - permanently defined spatial and temporal boundaries – will be published as such; - permanently defined lateral boundaries, with at least one vertical boundary (normally, upper limit) and temporal boundaries (as a time period) – lateral boundaries will be published as provided in the AIP, and the vertical and temporal boundaries will be provided from the dynamic data sources (normally a NOTAM); - restriction entirely defined exclusively through dynamic data source (NOTAM) – where all the data will be retrieved from the dynamic data source. When the data is retrieved from the dynamic data source (NOTAM), best effort is made to retrieve precise data about spatial and temporal limits from the Item E of the NOTAM (NOTAM text). However, as these data are provided in plain language (the only constraint being the use of ICAO abbreviations) it may be impossible to retrieve the precise 4D geometry through automation. In such a case, the 4D geometry for publication through CIS will be determined reading from the NOTAM Q field (area affected by the NOTAM), which will result in the 4D geometry that is more conservative (more constraining in spatial or temporal terms) than the actual airspace volume announced by the NOTAM.
Service Implementation and Structural Details	
Service interfaces **SWIM-SERV-240**	Refer to the following sections in ASTM F3548-21 for information about service interfaces. * A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description proposed with ED318
Service interface protocols and data format **SWIM-SERV-260**	Refer to the following sections in ASTM F3548-21 for information about service interface protocols and data format. * A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description proposed with ED318
Service operations **SWIM-SERV-270**	Refer to the following sections in ASTM F3548-21 for information about service operations. * A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description proposed with ED318
Service messages	Refer to the following sections in ASTM F3548-21 for information about messages exchanged by the service.

SWIM-SERV-280	* A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description proposed with ED318
Information Aspects of the Service	
Information definition (minimum) **SWIM-SERV-290**	Refer to the following sections in ASTM F3548-21 for information about the exchanged information service payload. * A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description proposed with ED318
Resources	
Machine-readable service interface definition **SWIM-SERV-320**	Refer to the following sections in ASTM F3548-21 for information about the exchanged information service payload. * A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description proposed with ED318
Abbreviations and acronyms **SWIM-SERV-350**	Refer to the following sections in ASTM F3548-21 for abbreviations and acronyms: * 3 - Terminology * A2 - Interoperability requirements and DSS testing * A3 - USS-DSS and USS-USS OpenAPI YAML Description proposed with ED318